

Conjecture 00000008761 is false under its stated definition

Ziang Ni (ziangni-sys)

10 October 2026

The English and Chinese statements define SK_1 as the kernel of $\det : GL(R) \rightarrow R^\times$ and assert its vanishing under a nilradical condition. We disprove this vanishing clause using precisely that definition.

Take the commutative field $R = \mathbb{Q}$. Its nilradical is zero: if $x^m = 0$ for $m > 0$, the absence of zero divisors implies $x = 0$. Consequently it also satisfies any nilpotent-annihilation condition on its nilradical: the zero ideal is annihilated already by the nilpotent zero element. Passing to the reduced ring changes nothing.

Consider the actual invertible matrix

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad U^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Direct multiplication gives $UU^{-1} = U^{-1}U = I_2$, and $\det U = 1$. However $U \neq I_2$, because its $(1, 2)$ entry is 1 rather than 0. Thus the determinant kernel is nontrivial.

This also works if $GL(R)$ denotes the stable general linear group. For every $n \geq 0$, the block diagonal matrix $U_n = \text{diag}(U, I_n)$ has determinant 1 and retains the same nonzero off-diagonal entry. Every standard stabilization is injective: $\text{diag}(A, 1) = \text{diag}(B, 1)$ implies $A = B$ by restricting to the original block. Hence the class of U never becomes the identity in the direct limit. It is a nonidentity element of the stable determinant kernel as well.

The claimed vanishing is therefore false. A false conjunct disproves the whole conjecture; the subsequent claim about noncommutative counterexamples need not be resolved.

Definition and formalization. The usual algebraic K -theory group SK_1 is instead a determinant kernel *after quotienting by elementary matrices*. Our matrix is elementary and becomes trivial in that quotient. We make no claim against a vanishing theorem for that different definition. The source explicitly defines the unquotiented kernel, and that is the statement refuted here.

The accompanying Lean project uses actual rational matrices and the general linear group type. Its final theorem proves that $\text{nilradical}(\mathbb{Q}) = 0$ and, for every n , there is a nonidentity invertible $(2 + n)$ -dimensional matrix of determinant one. The block indices $\text{Fin}(2) \sqcup \text{Fin}(n)$ express ordinary stabilization. All axiom audits use only Lean's standard logical axioms.